

SQL Server System Requirements

The following are the requirements to install the SQL Server when used with the latest available service pack:

It will support the below software and Operating Systems:

- Windows 10 (64-bit) version 1607 (10.0.14393) or later
- Windows 8.1 (64-bit)
- Windows Server 2019 (64-bit)
- Windows Server 2016 (64-bit)
- Windows Server 2012 R2 (64-bit)
- Windows Server 2012 (64-bit)
- Windows Server 2008 R2 (64-bit)
- .NET Framework

Before installing it on our system, we must have the following hardware:

- 4 GHz or faster x64 processor. For better use, it is recommended to have 2.0 GHz or faster.
- It requires a minimum of 512 MB RAM for express, and all other editions require 1 GB RAM. For better use, it is recommended 1 GB for express edition, and all other editions require 4 GB and should be increased when the database size increases.
- We must have a minimum of 6 GB space on the hard disk.

How to Download SQL Server Setup

We can use the following steps to download MS SQL Server in the Windows operating system:

Step 1: Go to the official page through this URL: <https://www.microsoft.com/en-in/sql-server/sql-server-downloads>.

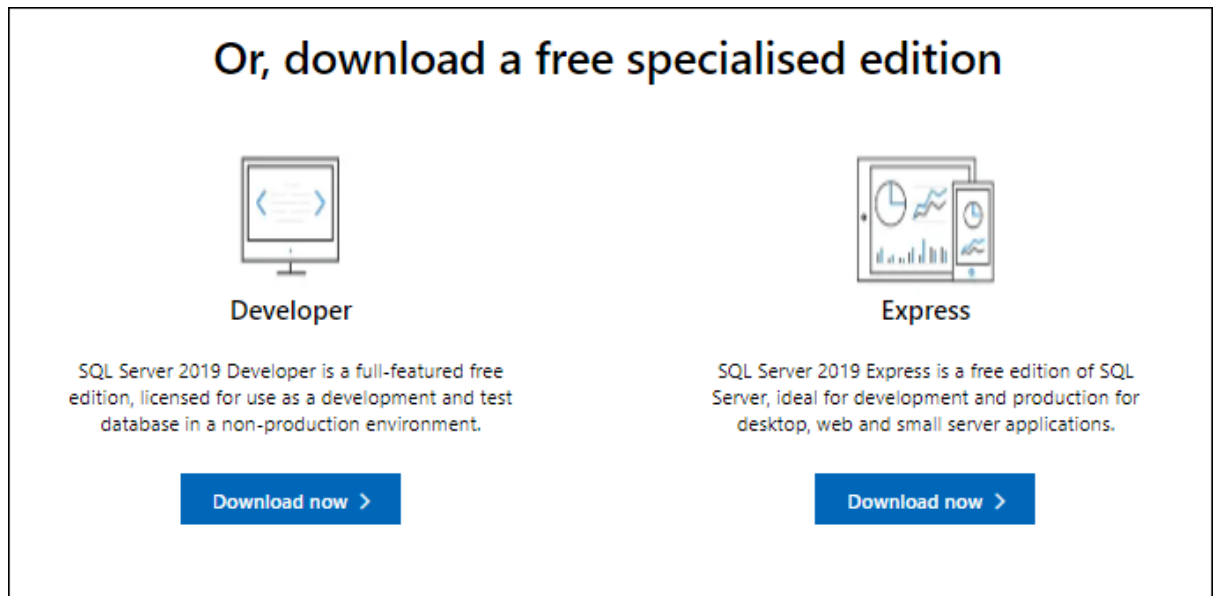
Microsoft offers two free editions that work with MS SQL Server:

- **Developer:** It has all of the MS SQL Server functionality, but we are unable to use it in a non-production environment as a development and test database. It's an ideal edition for the learning perspective.

- **Express:** It is also a free edition, but it comes with a limited set of features that is suitable for desktop, web, and small server application development and production.

Here, we are going to select the Express edition for installation on our system.

Step 2: Click on the "Download now" button. Immediately the SQL Server setup starts downloading on our system.

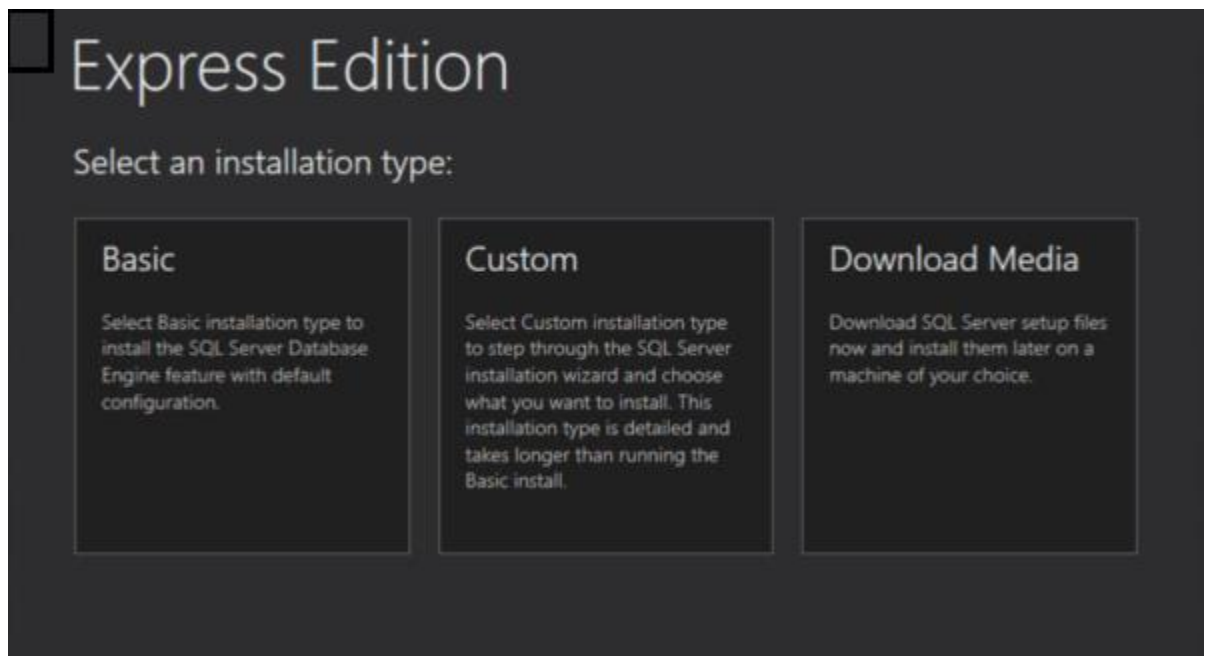


Step 3: Once the file has been downloaded, we will get the setup named "SQL2019-SSEI-Expr.exe."

How to Install SQL Server

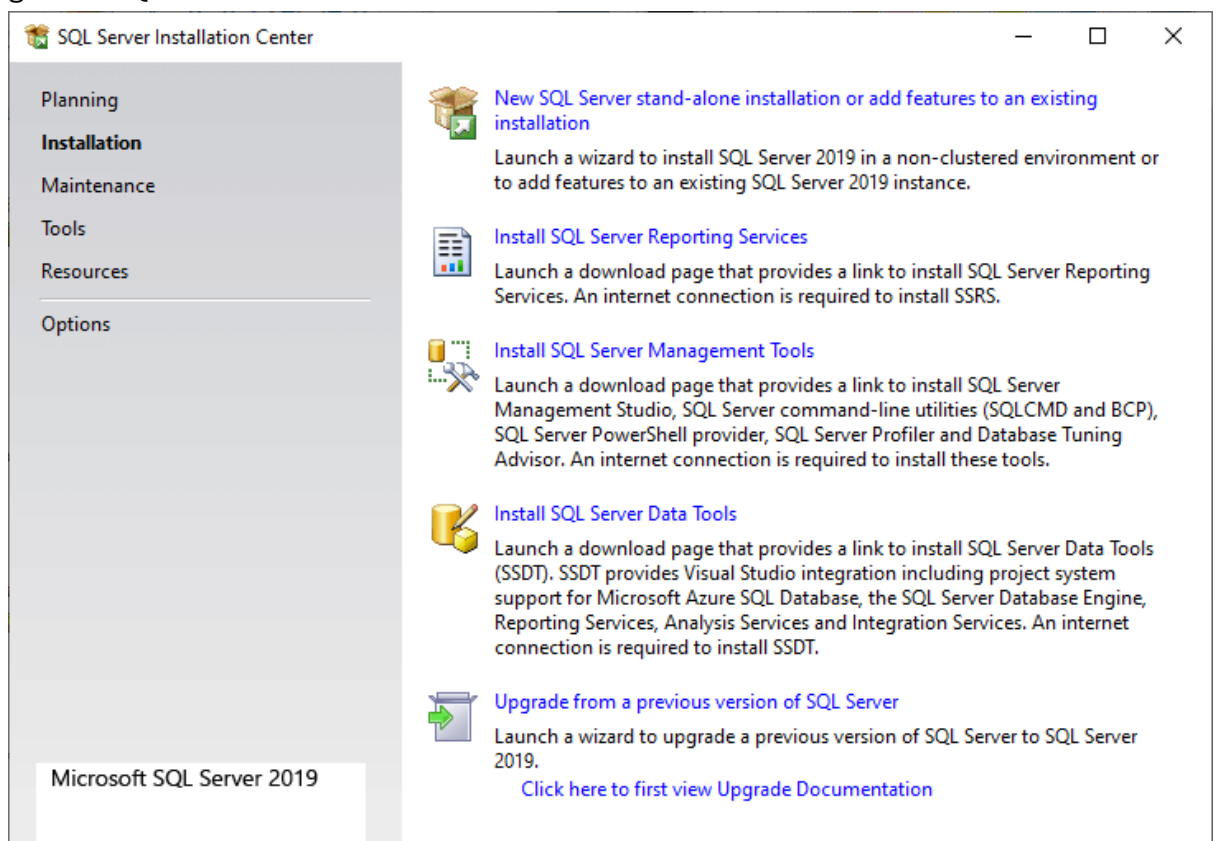
The following steps help us to install MS SQL Server in the Windows operating system:

Step 1: Double click on "SQL2019-SSEI-Expr.exe" setup file. We will get the below screen with three options: **Basic**, **Custom**, and **Download Media files**. Here, we will select the '**Basic**' option to install the basic version that includes all of the default configurations needed to learn Microsoft SQL Server.



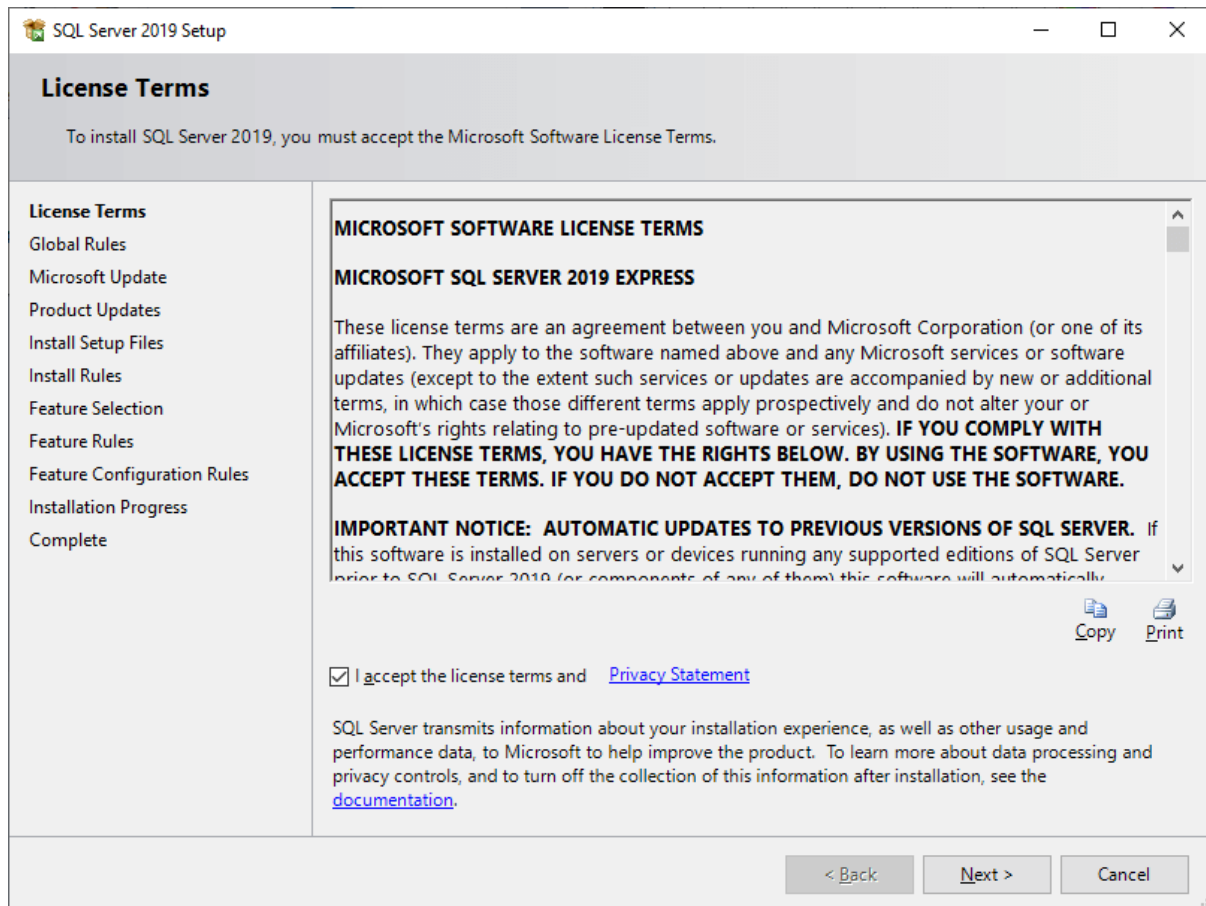
Step 2: The next wizard will ask you to accept the '**Microsoft Server License Terms**', click on the '**Accept**' button. Next, we will specify the SQL Server installation location; if necessary, we can change it to the desired place and click the **Install** button.

In the next wizard, we need to wait for the download process to complete installation packages. Once the installation is completed successfully, we will get the SQL Server Installation Center window similar to the below screen:



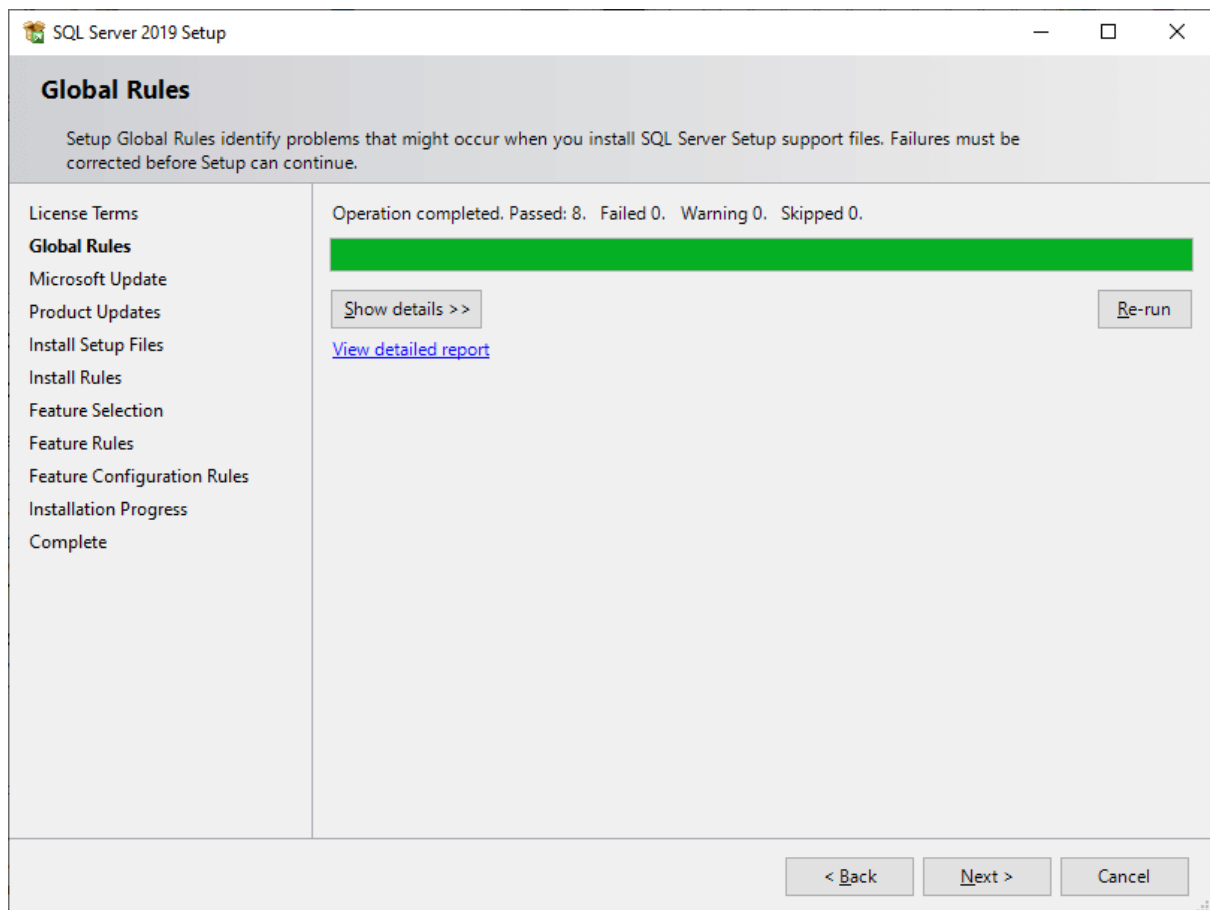
If you want to create a new SQL Server installation, select the **New SQL Server standalone installation or add features to an existing installation option**.

Step 3: On the next page, read the terms of the license agreement and then click on the Next button.

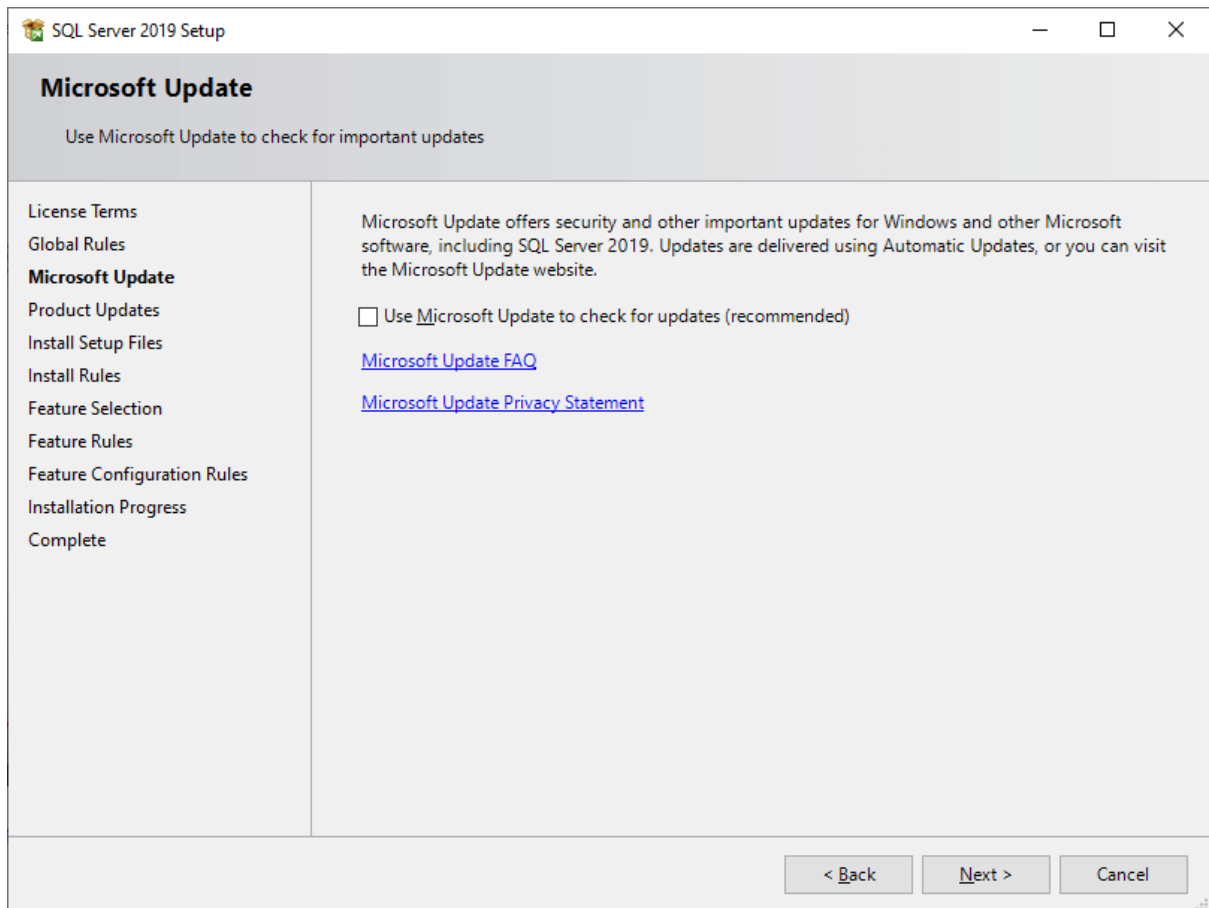


Step 4: In the next wizard, i.e., on the **Global Rules page**, if no rule error is found, the setup will automatically navigate to the **Product Updates page**.

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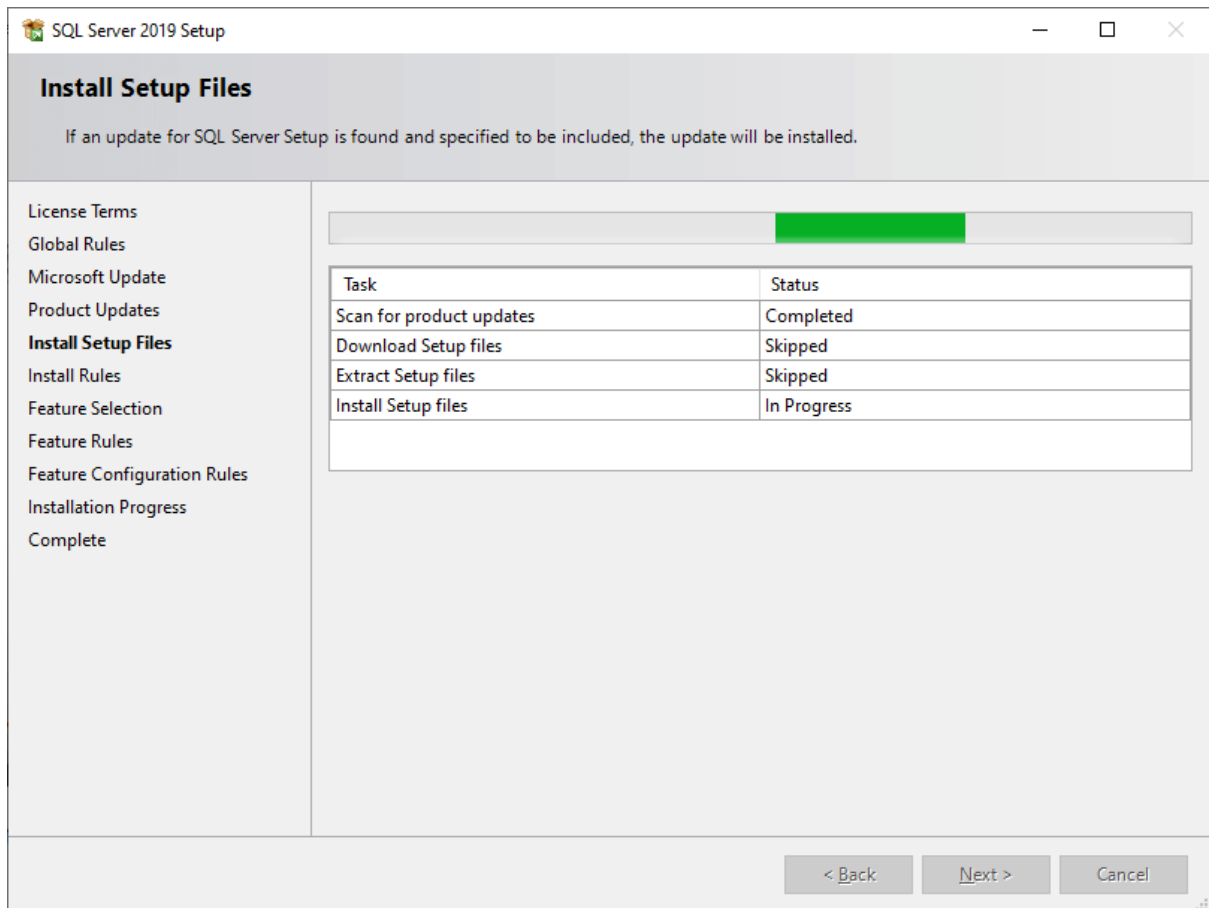


Step 5: The next page is a **Microsoft Update** page. When we click the check box, it will check the latest updates for all Microsoft products for changing the computer settings. After finishing the updates, click on Next.

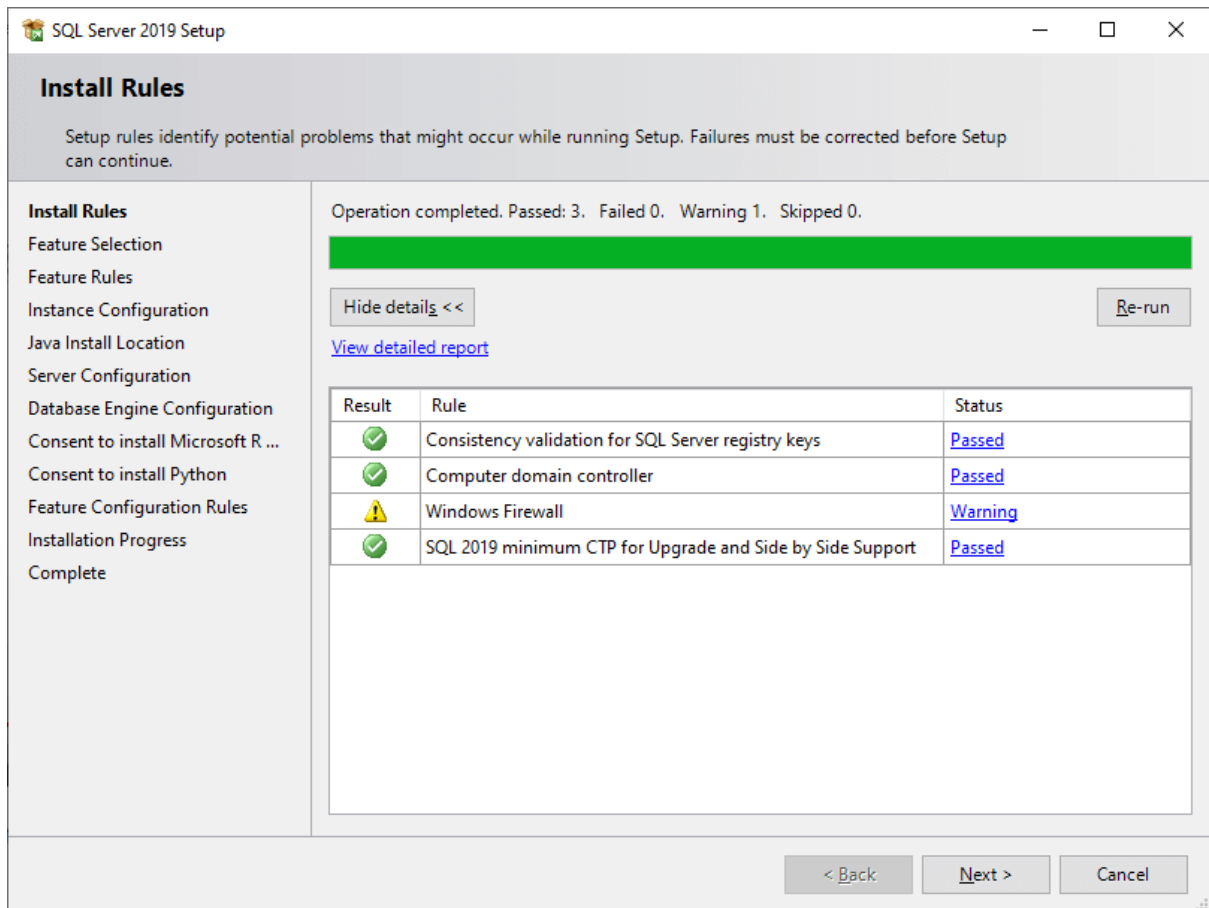


Step 6: The next wizard is the Product Updates page that shows the latest SQL Server software updates. If it does not find any product updates, the setup skips over this page and automatically navigates to the Install Setup Files page.

Step 7: The next wizard is the **Install Setup Files** page that displays the progress of the download, extract, and installation of the setup files. If it finds an update and we want to include these updates, it will be installed as well. If no update is found, the setup will automatically navigate to the next page.

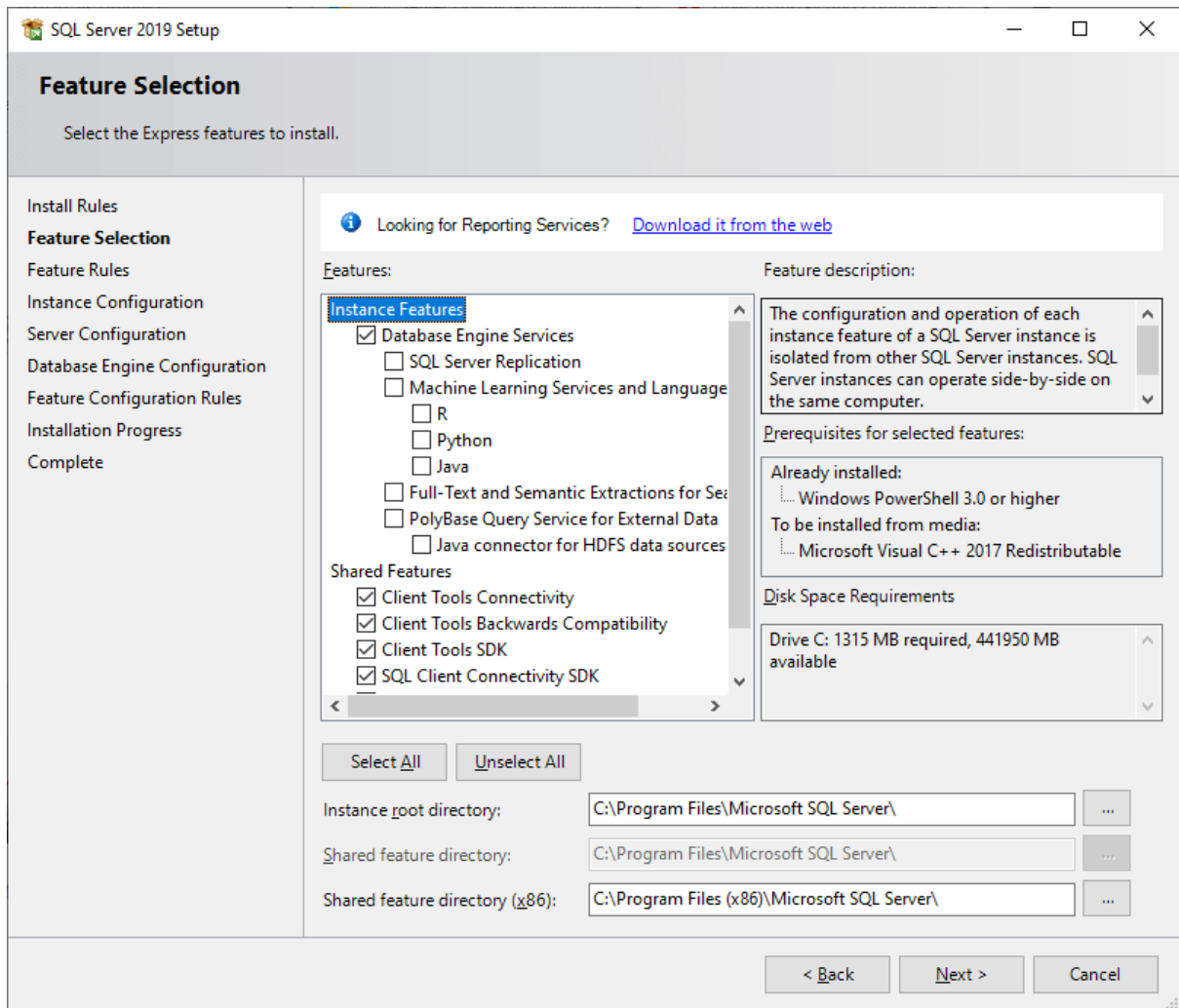


Step 8: The next wizard is the **Install Rules** page on which the setup tests for possible issues that can arise while running the setup. If it finds issues, select an item in the **Status** column for more detail. If no issues are found, click on Next.



Step 9: If we install this setup the first time on our system, it will skip the **Installation Type** page and navigate the **Feature Selection** page. If the setup is found on the system, we can choose whether to perform a new installation or add new features to it on the Installation Type page.

On the Feature Selection page, we need to choose the components for our installation. Suppose we want to install a new instance of SQL Server Database Engine, we'll choose **Database Engine Services**. Click Next to proceed.



Step 10: In the next wizard, we will specify the **name** and **instance ID** to create the new instance of SQL Server. Leave the default and click on Next to proceed.

SQL Server 2019 Setup

Instance Configuration

Specify the name and instance ID for the instance of SQL Server. Instance ID becomes part of the installation path.

Install Rules
Feature Selection
Feature Rules
Instance Configuration
Server Configuration
Database Engine Configuration
Feature Configuration Rules
Installation Progress
Complete

☐ Default instance
☒ Named instance:

Instance ID:

SQL Server directory: C:\Program Files\Microsoft SQL Server\MSSQL15.SQLEXPRESS

Installed instances:

Instance Name	Instance ID	Features	Edition	Version

< Back
Next >
Cancel

Step 11: The next wizard is the **Server Configuration** page, where we will specify the service account an

SQL Server 2019 Setup

Server Configuration

Specify the service accounts and collation configuration.

Install Rules
Feature Selection
Feature Rules
Instance Configuration
Server Configuration
Database Engine Configuration
Feature Configuration Rules
Installation Progress
Complete

Service Accounts Collation

Microsoft recommends that you use a separate account for each SQL Server service.

Service	Account Name	Password	Startup Type
SQL Server Database Engine	NT Service\MSSQL\$SQL...		Automatic
SQL Server Browser	NT AUTHORITY\LOCAL ...		Disabled

☐ Grant Perform Volume Maintenance Task privilege to SQL Server Database Engine Service

This privilege enables instant file initialization by avoiding zeroing of data pages. This may lead to information disclosure by allowing deleted content to be accessed.

[Click here for details](#)

< Back
Next >
Cancel

collation configuration. Here, we will set the startup type "**Automatic**" and click on Next to proceed.

Step 12: The next wizard is the **Database Engine Configuration**, where we will specify the authentication security mode, administrators, and data directories. Here we will select the following:

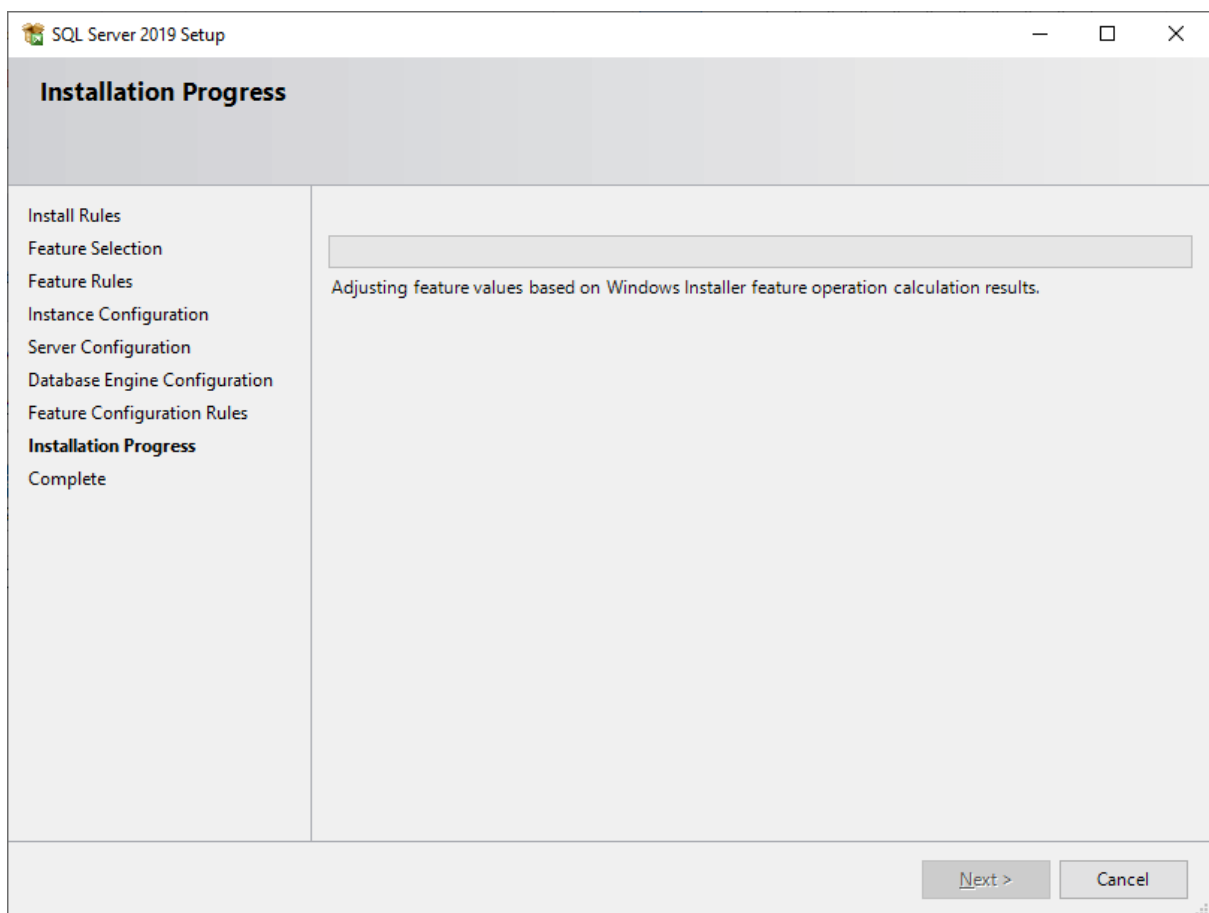
Authentication Mode

- Choose Mixed Mode
- Enter a STRONG password for the system administrator account

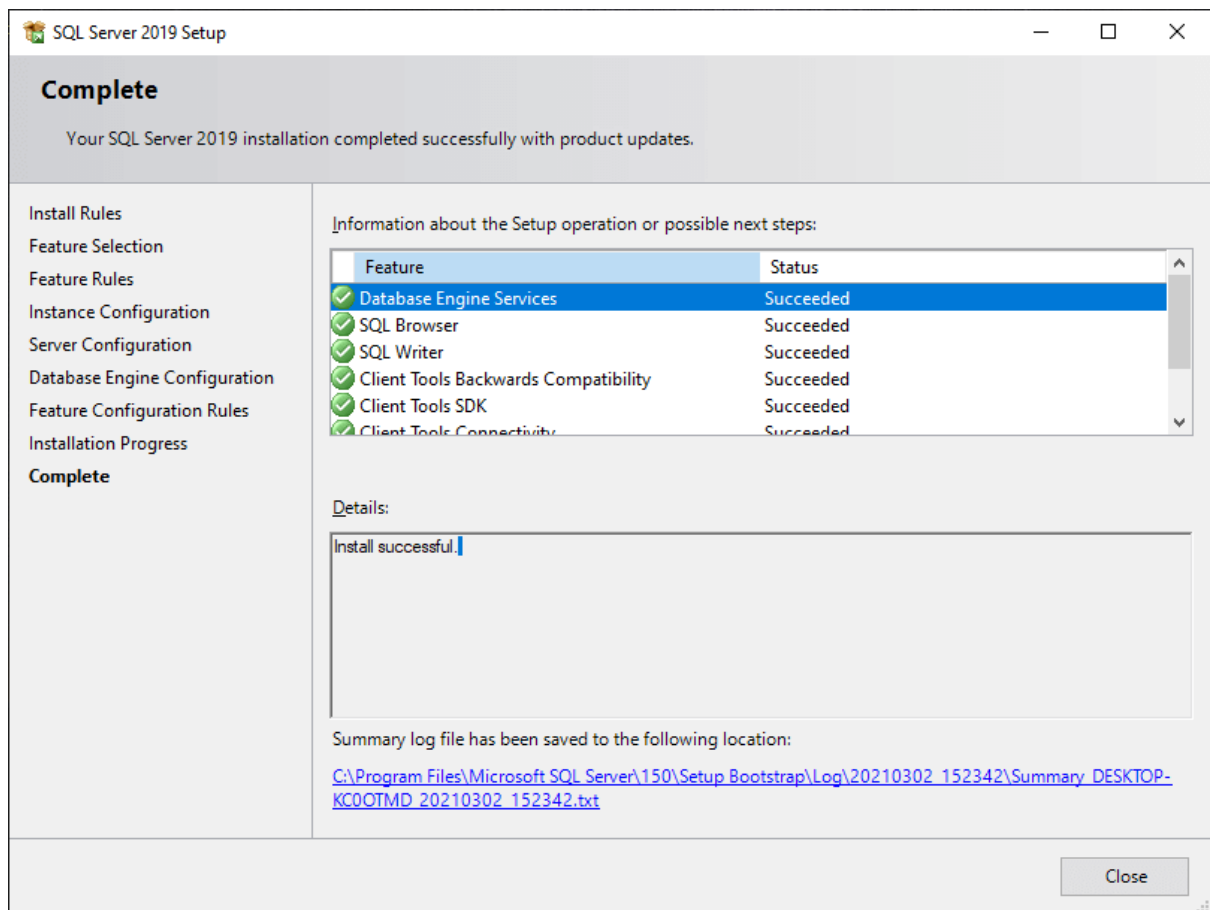
Specify SQL Server Administrators

- Add the local administrator account

Click on Next to proceed. The installation process begins.



Step 13: Once the installation is complete, click on the **Close** button to finish the setup.



Step 14: If you're instructed to restart your computer, do so now to complete the SQL Server setup.

SQL Server Management Studio (SSMS)

MS SQL Server Management Studio is a workstation or a client tool which is used to connect to and manage your SQL Server

SQL Server Management Studio (SSMS) is a **windows software or a client tool** used to connect and work with our SQL Server from a **graphical interface** instead of using the command line. Microsoft SQL Server 2005 launched the management studio to work with [SQL Server](#) and Azure SQL databases.

It allows DBAs and database developers to **configure, manage, and administer** all components within SQL Server. Its main functionality is to create databases and tables, execute SQL queries for inserting, updating, and deleting data, creating and managing stored procedures, triggers, views, and cursors. It also enables us to set privileges (securities) on databases and their objects.

SSMS also includes tools for deployment, database health monitoring, and reporting. It includes **SQL Profiler**, which allows us to examine the performance of our SQL databases. It's also possible to use it to schedule background work. If we want to connect to a remote SQL Server instance, we'll need this GUI tool or similar software. It is used by Administrators, Developers, Testers, etc. The latest version of SQL Server Management Studio is **SSMS 18.8 RC**. If we have installed a previous version, we just need to install **SSMS 18.8 upgrades**.

Backward Skip 10sPlay VideoForward Skip 10s

SSMS System Requirements

The following are the requirements to install the current release of SSMS when used with the latest available service pack:

It will support the below Operating Systems:

- Windows 10 (64-bit) version 1607 (10.0.14393) or later
- Windows 8.1 (64-bit)
- Windows Server 2019 (64-bit)
- Windows Server 2016 (64-bit)
- Windows Server 2012 R2 (64-bit)
- Windows Server 2012 (64-bit)
- Windows Server 2008 R2 (64-bit)

Before installing it on your system, we must have the following hardware:

- 8 GHz or faster x86 processor. For better use, it is recommended to have Dual-core.
- It requires a minimum of 2 GB RAM, but 4 GB is recommended (if it runs on a virtual machine, a minimum of 2.5 GB is required).
- We must have a minimum of 2 GB up to 10 GB space on the hard disk.

How to Download and Install SQL Server Management Studio (SSMS)

We can use the following steps to download and install Microsoft SQL Server Management Studio (SSMS) in the Windows system:

Step 1: Go to the official page by clicking on this [link](#) and click on [Download SQL Server Management Studio 18.08](#). Immediately the SSMS setup starts downloading on your system.

Applies to: ✓ SQL Server (all supported versions) ✓ Azure SQL Database ✓ Azure SQL Managed Instance ✓ Azure Synapse Analytics

SQL Server Management Studio (SSMS) is an integrated environment for managing any SQL infrastructure, from SQL Server to Azure SQL Database. SSMS provides tools to configure, monitor, and administer instances of SQL Server and databases. Use SSMS to deploy, monitor, and upgrade the data-tier components used by your applications, and build queries and scripts.

Use SSMS to query, design, and manage your databases and data warehouses, wherever they are - on your local computer, or in the cloud.

Download SSMS

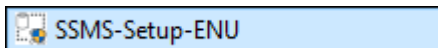


[Download SQL Server Management Studio \(SSMS\)](#)

SSMS 18.8 is the latest general availability (GA) version of SSMS. If you have a previous GA version of SSMS 18 installed, installing SSMS 18.8 upgrades it to 18.8.

Step 2: Once the file has been downloaded, it will be named "**SSMS-Setup-ENU.exe**." Now, double-click on it to continue.

Advertisement



If you see the prompt message "Do you want to allow the following to make a change this computer? Say **YES**."

Step 3: We will see a below window. Here, we can change the installation location and then click the '**Install**' button for installing SQL Server Management Studio (SSMS) on your device.



RELEASE 18.8

Microsoft SQL Server Management Studio with Azure Data Studio

Welcome. Click "Install" to begin.

Location:

C:\Program Files (x86)\Microsoft SQL Server Management Studio 18

Change

By clicking the "Install" button, I acknowledge that I accept the [Privacy Statement](#) and the License Terms for [SQL Server Management Studio](#) and [Azure Data Studio](#)

SQL Server Management Studio transmits information about your installation experience, as well as other usage and performance data, to Microsoft to help improve the product. To learn more about data processing and privacy controls, and to turn off the collection of this information after installation, see the [documentation](#)

Install

Close

Step 4: The installation process will start. We will see the progress of the packages and the overall progress on the below screen. Since it downloads the necessary packages from the internet, the installation can take some time.



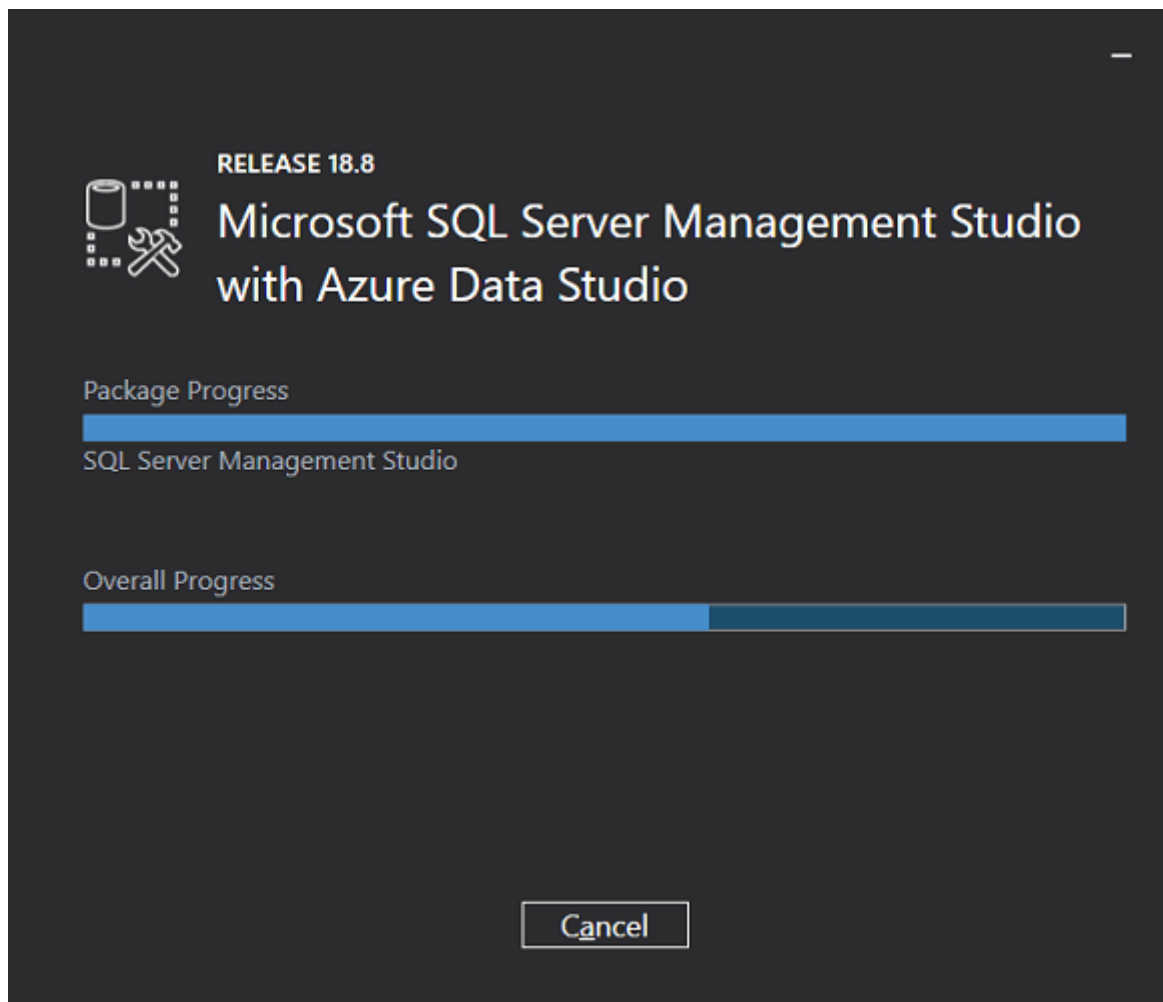
RELEASE 18.8

Microsoft SQL Server Management Studio with Azure Data Studio

Loading packages. Please wait...



Cancel



Step 5: When SSMS Setup is finished, it will show the screen below with the "**Setup Completed**" message. It's possible that we may prompt us to restart our computer.



RELEASE 18.8

Microsoft SQL Server Management Studio with Azure Data Studio

Restart required in order to complete setup.

All specified components have been installed successfully.

The computer needs to be restarted before setup can continue.

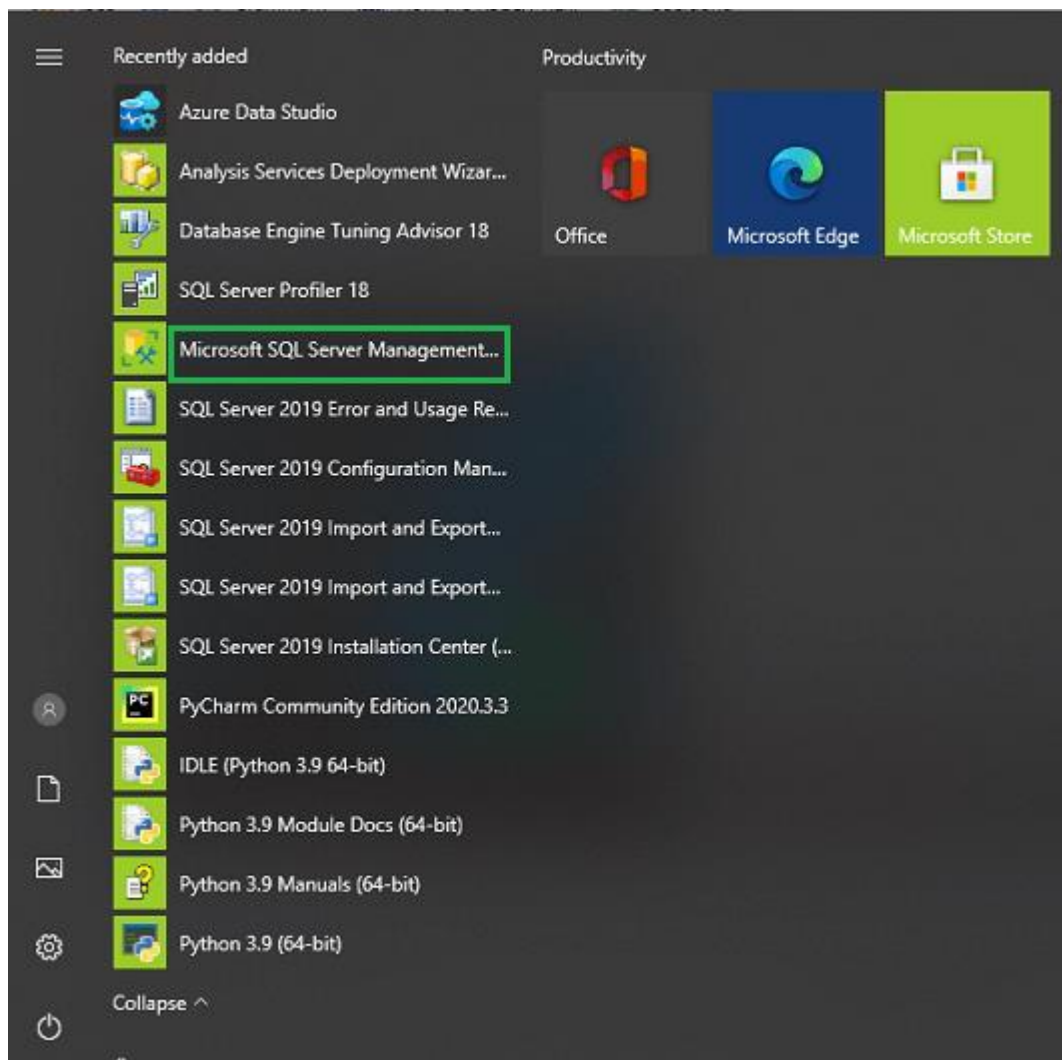
Restart

Close

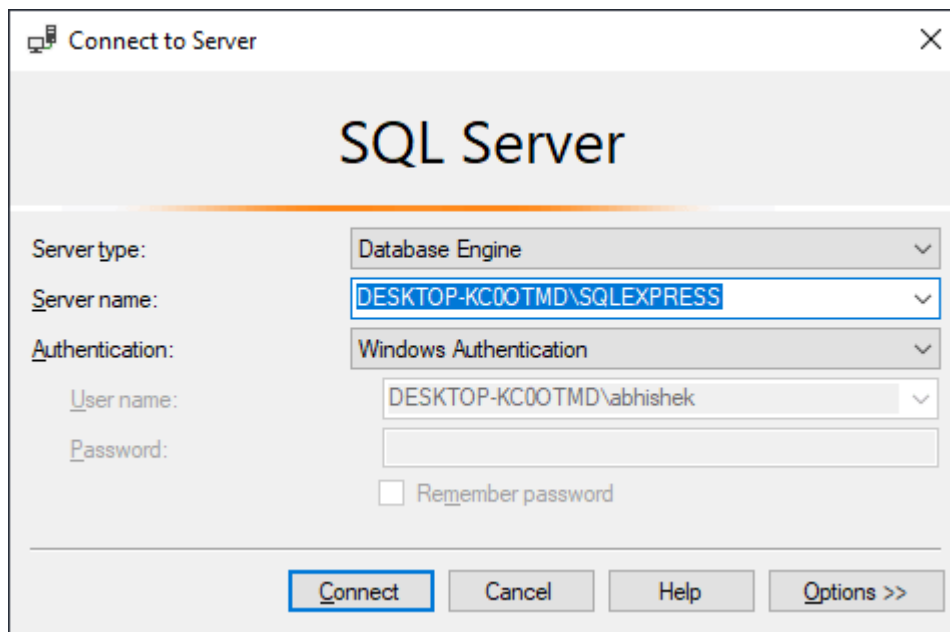
How to access SQL Server Management Studio

We can use the following steps to access SQL Server Management Studio (SSMS):

Go to Start Menu>Programs>Microsoft SQL Server Tools 18> Microsoft SSMS 18.



The '**Connect to Server**' dialogue screen will appear in front of us when we start SSMS. Set the **Server name** and **Authentication type** of the SQL Server we want to connect to on this screen, then click the **Connect** button to be connected to '**Data Management Studio**'. Here the server name defaults to the name chosen during the installation of the MS SQL server.



Let us understand each field of the above screen in detail:

Server type: It is a dropdown menu that allows us to choose one of four MS SQL services options. Here we'll be working on the '**Database Engine**' that enables us to create and work with databases. Some of the examples of other server types are Analysis, Reporting, and Integration Services.

Server name: It is the server's name where [MS SQL Server](#) is installed, and we will need to connect with that server. In most cases, we refer to the server name as "Machine name/Instance." Here we had given the Instance name to the SQL Server instance during installation.

Authentication: If we use "**Windows Authentication**" during SQL Server's installation, this option is set to "Windows Authentication." Authentication will be set to "**SQL Server Installation**" if we have chosen "**Mixed Mode** (Windows Authentication & Windows Authentication)."

User name\Password: These two fields will be needed when the Authentication is set to other than "Windows Authentication," such as "SQL Server Installation."

How to access SQL Server Management studio using Command line?

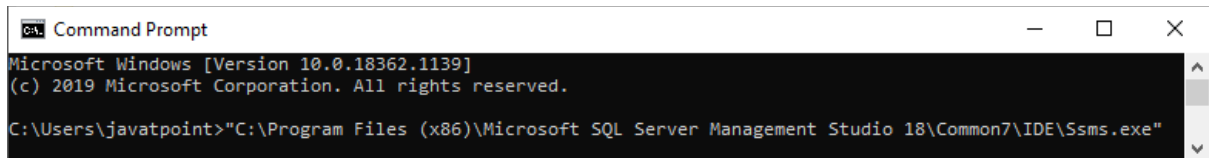
We can also use the Window Command Line tool to launch SQL Server Management Studio. We can do this by using the complete path to ssms.exe. The default location and file name are as follows:

1. C:\Program Files (x86)\Microsoft SQL Server Management Studio 18\Common7\IDE
2. Exe name: ssms.exe

The steps to access Management Studio through the command line are as follows:

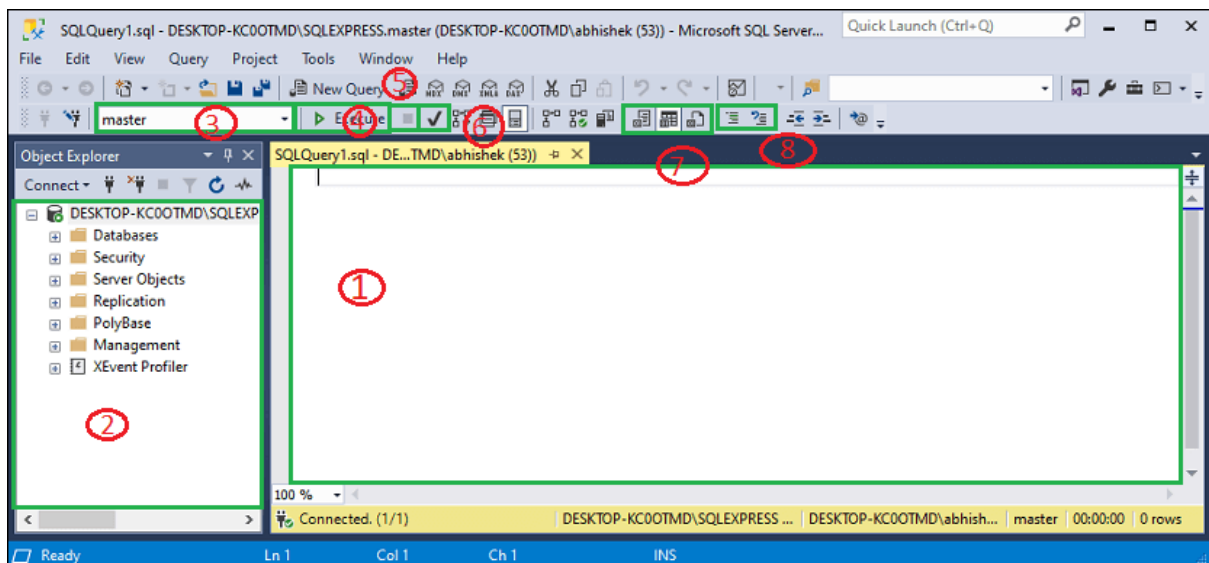
- Type CMD into the Search box to open Command Prompt.
- Next, we need to type the complete path followed by ssms.exe and press Enter.

Immediately, we will see the 'Connect to Server' screen similar to the one we have described above.



Introduction to Data Management Studio IDE

The Data Management Studio launch screen will look like below. We will discuss each section in detail.



1. Query Editor

This section is used to write all of our queries. MS SQL Server offers interactive recommendations for tables, columns, and other items for easy queries creations and a variety of other features.

Advertisement

2. Object Explorer

The Object Explorer shows the database objects contained on the server in a tree format. For quick reference, this section lists all of the Databases, Security, and Server Objects. We can see each object's components by simply pressing the + icon to the left of the object to expand it.

3. Databases Selection Dropdown

It is a dropdown option that allows the user to choose the database to run our queries.

4. Execute button

It is an execution button that allows the user to execute the query and return the results.

5. Cancel Executing Query

It is a button that is used to stop the execution of the query.

6. Parse

It is a button used to validate the query's syntax but doesn't check the database objects.

7. Changing query result destination

This button group allows us to set the query result's destination. There are three options available:

- Results to Grid
- Results to Text
- Results to File

8. Comment/Uncomment out the Selected Lines

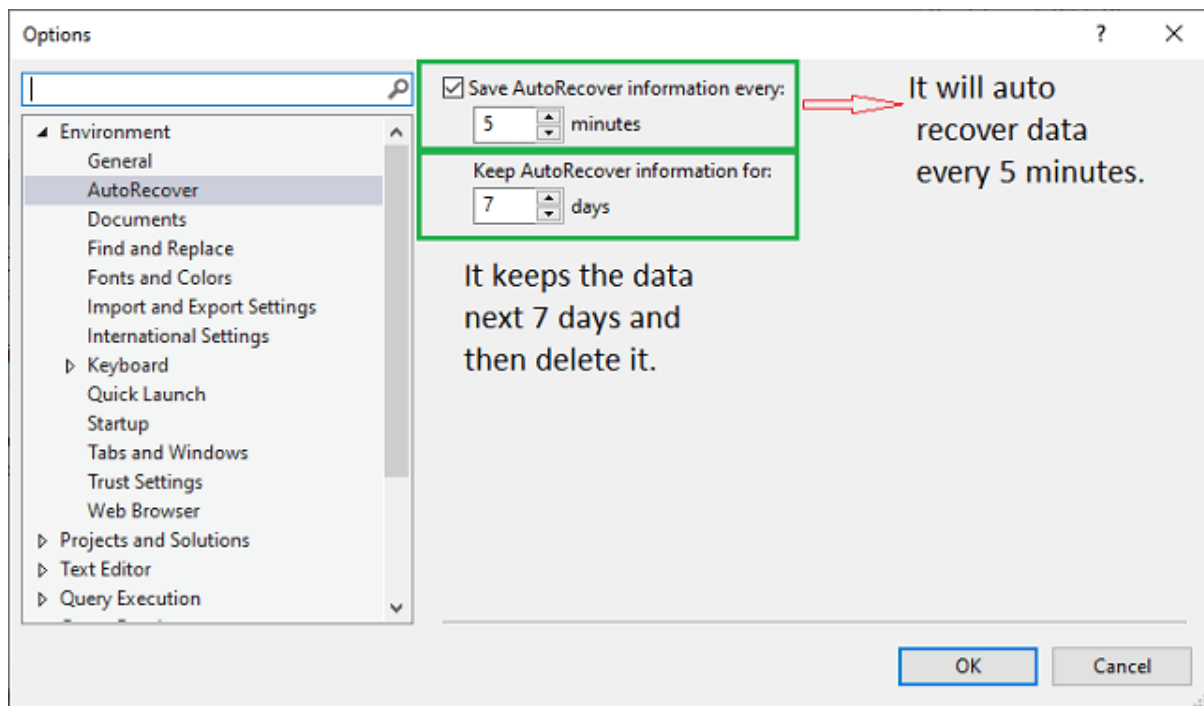
These buttons allow us to comment or uncomment the lines of code we have chosen. For example, to see the line numbers of the code in the query editor.

Important points to remember

The following points must be useful while using the SQL Server Management Studio:

- SSMS is a standalone application that doesn't refer to any particular SQL Server version. **For example**, we may use SMMS Version 18 with SQL Server 2017 and SQL Server 2016.
- If we write the larger code, it reduces the readability. In that case, we need to make use of comments for better readability. We can comment out a line, just putting "--" in front of it.
- A sudden shutdown or a device failure may result in data loss. We can minimize the data loss by setting the '**AutoRecover**' option checked. We can also configure the AutoRecover data time-period and the number of days this information is stored before being deleted. We can set the AutoRecover option as below:

Tools> Options>Environment>AutoRecover



SSMS Versions and Updates

The first version of SSMS was released with SQL Server 2005. With SQL Server 2008, SQL Server 2012, and SQL Server 2016, SSMS is an integral part of SQL Server. Microsoft began the first numerical versioning sequence in 2016, with the SSMS version number 16.3. Version numbers for all series include 16.x, 17.x, and 18.0.

The latest series, which is 18.0, have the following versions: 18.0 (Preview 4), 18.0 (Preview 5), and 18.0 (Preview 6). As of March 2021, the current version of SQL Server Management Studio is SSMS 18.8 RC.